

(Some of these problems are labeled with a 3, but relate to material from chapter 4)

3.95 Are the following instructions for diluting a 10.0 *M* solution to a 1.00 *M* solution correct: "Take 100.0 mL of the 10.0 *M* solution and add 900.0 mL water"? Explain.

3.96 Calculate each of the following quantities:

- (a) Grams of solute in 185.8 mL of 0.267 *M* calcium acetate
- (b) Molarity of 500. mL of solution containing 21.1 g of potassium iodide
- (c) Moles of solute in 145.6 L of 0.850 *M* sodium cyanide

3.100 Calculate each of the following quantities:

- (a) Molarity of a solution prepared by diluting 37.00 mL of 0.250 *M* potassium chloride to 150.00 mL
- (b) Molarity of a solution prepared by diluting 25.71 mL of 0.0706 *M* ammonium sulfate to 500.00 mL
- (c) Molarity of sodium ion in a solution made by mixing 3.58 mL of 0.348 *M* sodium chloride with 500. mL of 6.81×10^{-2} *M* sodium sulfate (assume volumes are additive)

3.102 A sample of concentrated nitric acid has a density of 1.41 g/mL and contains 70.0% HNO_3 by mass.

- (a) What mass of HNO_3 is present per liter of solution?
- (b) What is the molarity of the solution?

3.114 Narceine is a narcotic in opium that crystallizes from solution as a hydrate that contains 10.8 mass % water and has a molar mass of 499.52 g/mol. Determine *x* in narceine·*x*H₂O.

3.134 Assuming that the volumes are additive, what is the concentration of KBr in a solution prepared by mixing 0.200 L of 0.053 *M* KBr with 0.550 L of 0.078 *M* KBr?

3.161 Cocaine ($\text{C}_{17}\text{H}_{21}\text{O}_4\text{N}$) is a natural substance found in coca leaves, which have been used for centuries as a local anesthetic and stimulant. Illegal cocaine arrives in the United States either as the pure compound or as the hydrochloride salt ($\text{C}_{17}\text{H}_{21}\text{O}_4\text{NHCl}$). At 25°C, the salt is very soluble in water (2.50 kg/L), but cocaine is much less so (1.70 g/L). (a) What is the maximum amount (in g) of the salt that can dissolve in 50.0 mL of water? (b) If the solution in part (a) is treated with NaOH, the salt is converted to cocaine. How much additional water (in L) is needed to dissolve it?

4.2 What types of substances are most likely to be soluble in water?

4.3 What must be present in an aqueous solution for it to conduct an electric current? What general classes of compounds form solutions that conduct?

4.10 State whether each of the following substances is likely to be very soluble in water. Explain.

- (a) Benzene, C_6H_6
- (b) Sodium hydroxide
- (c) Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$
- (d) Potassium acetate

4.12 State whether an aqueous solution of each of the following substances conducts an electric current. Explain your reasoning.

- (a) Cesium iodide
- (b) Hydrogen bromide

4.18 How many moles and numbers of ions of each type are present in the following aqueous solutions?

- (a) 100. mL of 2.45 *M* aluminum chloride
- (b) 1.80 L of a solution containing 2.59 g lithium sulfate/L
- (c) 225 mL of a solution containing 1.68×10^{22} formula units of potassium bromide per liter

4.20 How many moles of H^+ ions are present in the following aqueous solutions?

- (a) 1.40 L of 0.25 *M* perchloric acid
- (b) 1.8 mL of 0.72 *M* nitric acid
- (c) 7.6 L of 0.056 *M* hydrochloric acid

4.24 Which ions do not appear in a net ionic equation? Why?

4.29 Complete the following precipitation reactions with balanced molecular, total ionic, and net ionic equations:

- (a) $\text{Hg}_2(\text{NO}_3)_2(aq) + \text{KI}(aq) \longrightarrow$
- (b) $\text{FeSO}_4(aq) + \text{Ba}(\text{OH})_2(aq) \longrightarrow$

4.31 When each of the following pairs of aqueous solutions is mixed, does a precipitation reaction occur? If so, write balanced molecular, total ionic, and net ionic equations:

- (a) Sodium nitrate + copper(II) sulfate
- (b) Ammonium iodide + silver nitrate

4.33 When each of the following pairs of aqueous solutions is mixed, does a precipitation reaction occur? If so, write balanced molecular, total ionic, and net ionic equations:

- (a) Potassium chloride + iron(II) nitrate
- (b) Ammonium sulfate + barium chloride

4.35 If 35.0 mL of lead(II) nitrate solution reacts completely with excess sodium iodide solution to yield 0.628 g of precipitate, what is the molarity of lead(II) ion in the original solution?

4.37 The mass percent of Cl^- in a seawater sample is determined by titrating 25.00 mL of seawater with AgNO_3 solution, causing a precipitation reaction. An indicator is used to detect the end point, which occurs when free Ag^+ ion is present in solution after all the Cl^- has reacted. If 43.63 mL of 0.3020 *M* AgNO_3 is required to reach the end point, what is the mass percent of Cl^- in the seawater (*d* of seawater = 1.04 g/mL)?

4.45 Complete the following acid-base reactions with balanced molecular, total ionic, and net ionic equations:

- (a) Potassium hydroxide(*aq*) + hydriodic acid(*aq*) $\longrightarrow$
- (b) Ammonia(*aq*) + hydrochloric acid(*aq*) $\longrightarrow$

4.49 If 25.98 mL of a standard 0.1180 *M* KOH solution reacts with 52.50 mL of CH_3COOH solution, what is the molarity of the acid solution?

(Note: CH_3COOH is also known as $\text{HC}_2\text{H}_3\text{O}_2$)

4.58 In which of the following equations does sulfuric acid act as an oxidizing agent? In which does it act as an acid? Explain.

- (a) $4\text{H}^+(aq) + \text{SO}_4^{2-}(aq) + 2\text{NaI}(s) \longrightarrow$
 $2\text{Na}^+(aq) + \text{I}_2(s) + \text{SO}_2(g) + 2\text{H}_2\text{O}(l)$
- (b) $\text{BaF}_2(s) + 2\text{H}^+(aq) + \text{SO}_4^{2-}(aq) \longrightarrow$
 $2\text{HF}(aq) + \text{BaSO}_4(s)$

4.60 Give the oxidation number of carbon in the following:

- (a) CF_2Cl_2 (b) $\text{Na}_2\text{C}_2\text{O}_4$ (c) HCO_3^- (d) C_2H_6

4.62 Give the oxidation number of nitrogen in the following:

- (a) NH_2OH (b) N_2H_4 (c) NH_4^+ (d) HNO_2

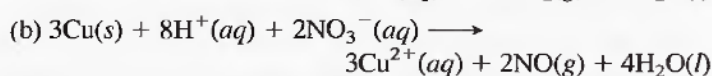
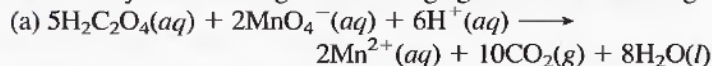
4.64 Give the oxidation number of arsenic in the following:

- (a) AsH_3 (b) H_3AsO_4 (c) AsCl_3

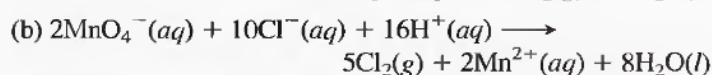
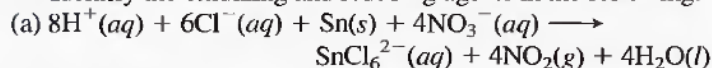
4.66 Give the oxidation number of manganese in the following:

- (a) MnO_4^{2-} (b) Mn_2O_3 (c) KMnO_4

4.68 Identify the oxidizing and reducing agents in the following:

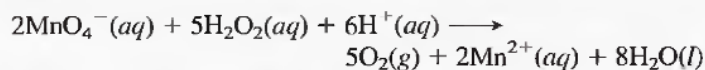


4.70 Identify the oxidizing and reducing agents in the following:



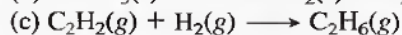
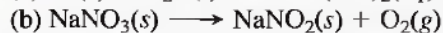
4.76 The active agent in many hair bleaches is hydrogen peroxide.

The amount of hydrogen peroxide in 13.8 g of hair bleach was determined by titration with a standard potassium permanganate solution:

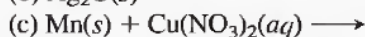
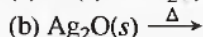
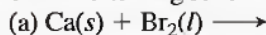


- (a) How many moles of MnO_4^- were required for the titration if 43.2 mL of 0.105 M KMnO_4 was needed to reach the end point? (b) How many moles of H_2O_2 were present in the 13.8-g sample of bleach? (c) How many grams of H_2O_2 were in the sample? (d) What is the mass percent of H_2O_2 in the sample? (e) What is the reducing agent in the redox reaction?

4.83 Balance each of the following redox reactions and classify it as a combination, decomposition, or displacement reaction:

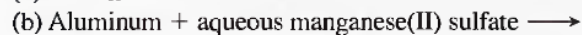


4.87 Predict the product(s) and write a balanced equation for each of the following redox reactions:



(Note: Δ means that heat is applied to the reactants)

4.91 Predict the product(s) and write a balanced equation for each of the following redox reactions:



4.93 How many grams of O_2 can be prepared from the thermal decomposition of 4.27 kg of HgO ? Name and calculate the mass (in kg) of the other product.

4.95 In a combination reaction, 1.62 g of lithium is mixed with 6.00 g of oxygen. (a) Which reactant is present in excess? (b) How many moles of product are formed? (c) After reaction, how many grams of each reactant and product are present?

4.97 A mixture of KClO_3 and KCl with a mass of 0.900 g was heated to produce O_2 . After heating, the mass of residue was 0.700 g. Assuming all the KClO_3 decomposed to KCl and O_2 , calculate the mass percent of KClO_3 in the original mixture.

4.99 Before arc welding was developed, a displacement reaction involving aluminum and iron(III) oxide was commonly used to produce molten iron (the thermite process). This reaction was used, for example, to connect sections of iron railroad track. Calculate the mass of molten iron produced when 1.00 kg of aluminum reacts with 2.00 mol of iron(III) oxide.

4.100 Iron reacts rapidly with chlorine gas to form a reddish brown, ionic compound (A), which contains iron in the higher of its two common oxidation states. Strong heating decomposes compound A to compound B, another ionic compound, which contains iron in the lower of its two oxidation states. When compound A is formed by the reaction of 50.6 g of Fe and 83.8 g of Cl_2 and then heated, how much compound B forms?

4.104 When either a mixture of NO and Br_2 or pure nitrosyl bromide (NOBr) is placed in a reaction vessel, the product mixture contains NO , Br_2 , and NOBr . Explain.

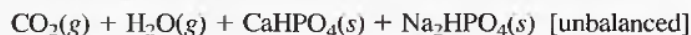
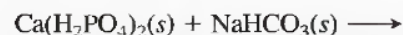
4.106 Nutritional biochemists have known for decades that acidic foods cooked in cast-iron cookware can supply significant amounts of dietary iron (ferrous ion).

(a) Write a balanced net ionic equation, with oxidation numbers, that supports this fact.

(b) Measurements show an increase from 3.3 mg of iron to 49 mg of iron per $\frac{1}{2}$ -cup (125-g) serving during the slow preparation of tomato sauce in a cast-iron pot. How many ferrous ions are present in a 26-oz (737-g) jar of the tomato sauce?

4.119 Sodium peroxide (Na_2O_2) is often used in self-contained breathing devices, such as those used in fire emergencies, because it reacts with exhaled CO_2 to form Na_2CO_3 and O_2 . How many liters of respired air can react with 80.0 g of Na_2O_2 if each liter of respired air contains 0.0720 g of CO_2 ?

4.126 Calcium dihydrogen phosphate, $\text{Ca}(\text{H}_2\text{PO}_4)_2$, and sodium hydrogen carbonate, NaHCO_3 , are ingredients of baking powder that react with each other to produce CO_2 , which causes dough or batter to rise:



If the baking powder contains 31% NaHCO_3 and 35% $\text{Ca}(\text{H}_2\text{PO}_4)_2$ by mass:

(a) How many moles of CO_2 are produced from 1.00 g of baking powder?

(b) If 1 mol of CO_2 occupies 37.0 L at 350°F (a typical baking temperature), what volume of CO_2 is produced from 1.00 g of baking powder?